

Training a domain-robust $\pi 0.5$ policy

FARM UF850 · overnight 4×H100 run

2026-05-30

GOAL Train the best deployable bottle-manipulation policy possible overnight on 4 GPUs.

DIAGNOSIS The deployed full fine-tune fails live mainly because of a DOMAIN SHIFT – demos were recorded in one room (living-room background, fixed camera) and the policy runs in another (plywood wall, repositioned camera, a person in frame). No fine-tune architecture fixes a domain shift; the standard training-time mitigation is DOMAIN RANDOMIZATION, plus preserving $\pi 0.5$'s robust pretrained vision priors instead of erasing them with a full fine-tune.

APPROACH A controlled 2×2 ablation, all GSE (base-preserving), all the proven 4-GPU recipe: (visual aug: default | HEAVY domain-randomization) × (prompt paraphrase: off | on) Heavy aug adds hue / channel-gamma / grayscale / blur / stronger crop-rotate on top of openpi's stock jitter. Models are scored both on the ORIGINAL frames (fit) and under test-time perturbations that mimic the room change (robustness – the metric that matters).

CAVEAT Synthetic perturbations are a PROXY for the real plywood-room shift. The only true test is live deployment / collecting demos in the target environment. Read the robustness numbers as 'how gracefully does the policy hold up when the scene appearance changes', not as a guaranteed live success rate.

Executive summary

Joint action error (deg), open-loop on held episodes:

model	clean	perturb	Δ degr	room-combo
full-FT (deployed)	0.73	1.44	+0.71	1.70
GSE (vanilla)	1.01	1.94	+0.93	2.03
GSE + prompt-aug	1.05	2.07	+1.03	2.18
GSE + heavy-aug	1.30	1.69	+0.39	1.32
GSE + aug + prompt (FLAGSHIP)	1.33	1.71	+0.38	1.31

READING IT (honest, not oversold):

- The full fine-tune fits the ORIGINAL frames tightest (0.7° vs $\sim 1.3^\circ$) and stays ahead on MILD shifts – it memorised the task in the training room.
- Heavy augmentation buys STABILITY: smallest degradation under shift ($\Delta \approx +0.4^\circ$ vs the full FT's $+0.7^\circ$), and on the realistic stacked 'room-change combo' the aug models hold $\sim 1.3^\circ$ while the deployed full FT degrades to $\sim 1.7^\circ$. The 2x2 isolates visual aug as THE lever ($2.0^\circ \rightarrow 1.3^\circ$).
- Prompt-aug didn't help this (visual) metric; its value is language generalization, which this action-error eval doesn't probe.

→ For a genuine room change (the actual live-failure mode), the augmented flagship is the better bet. This is an offline PROXY – confirm on the arm.

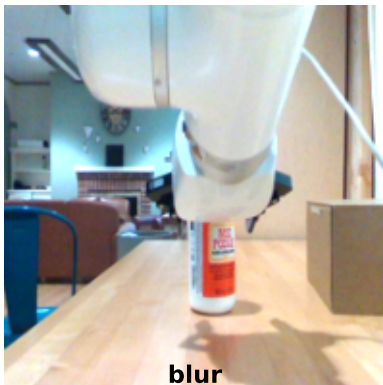
Pages follow: headline robustness, clean-vs-robust gap, the 2x2 ablation, clean fit, per-condition heatmap, per-task, loss curves, latency, conclusions.

What the robustness eval tests — one training frame under each domain-shift perturbation

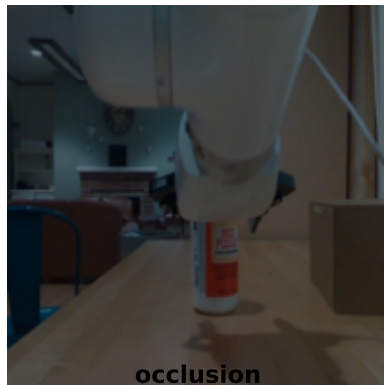
clean



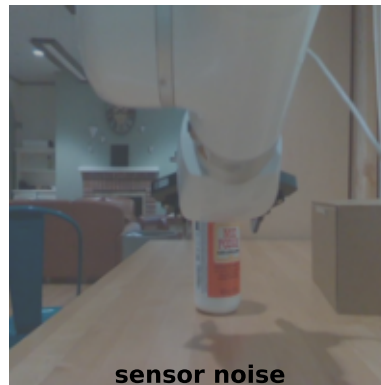
bright



dark



low contrast



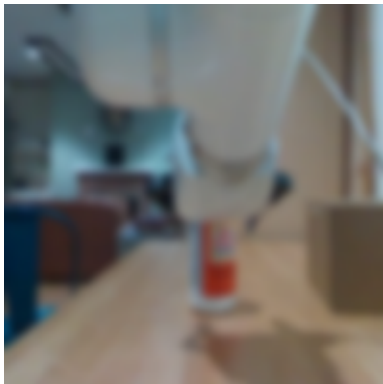
hue shift



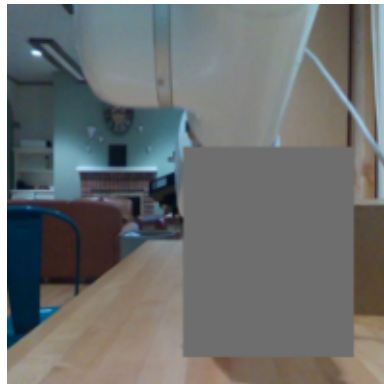
desaturate



blur



occlusion



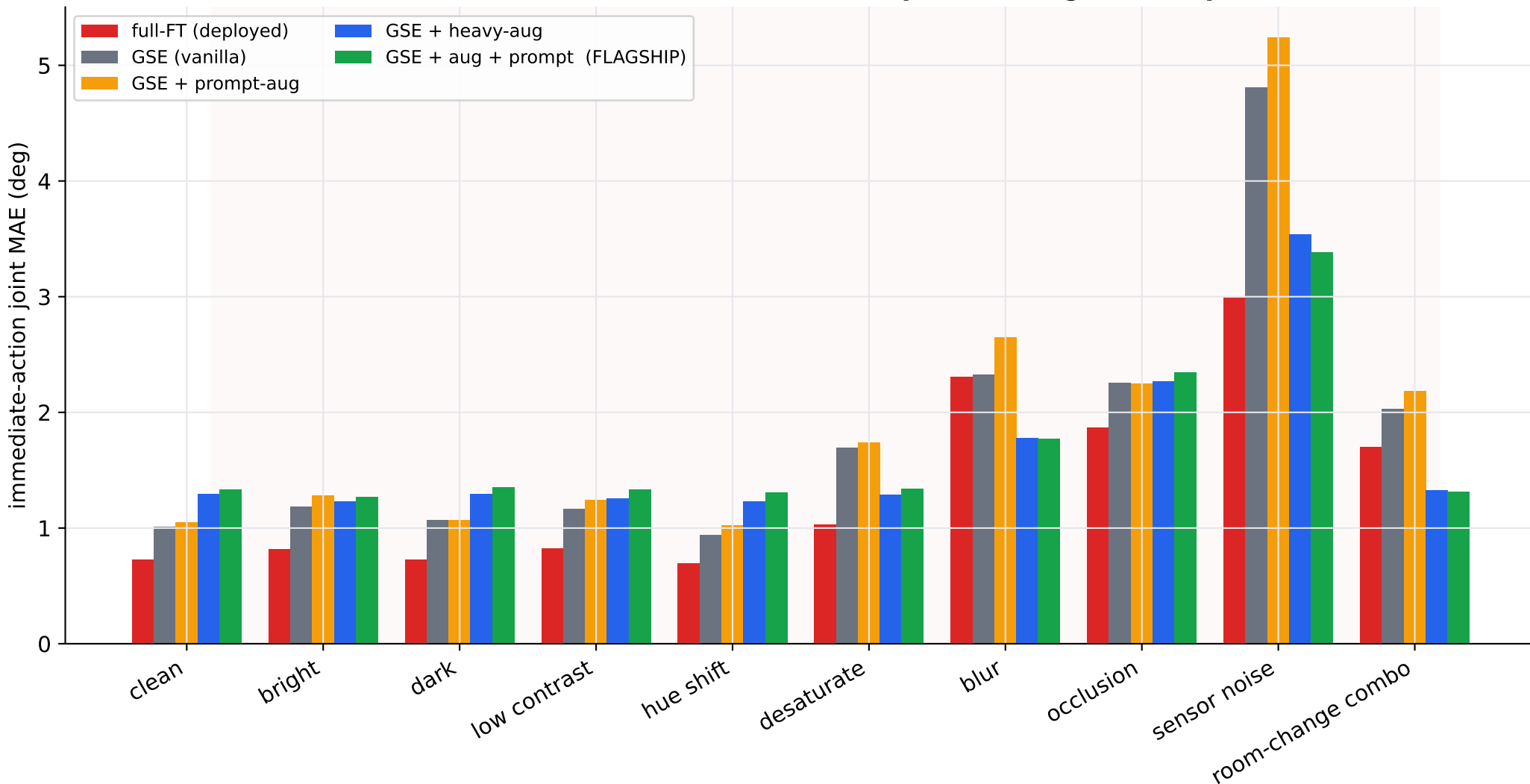
sensor noise



room-change combo

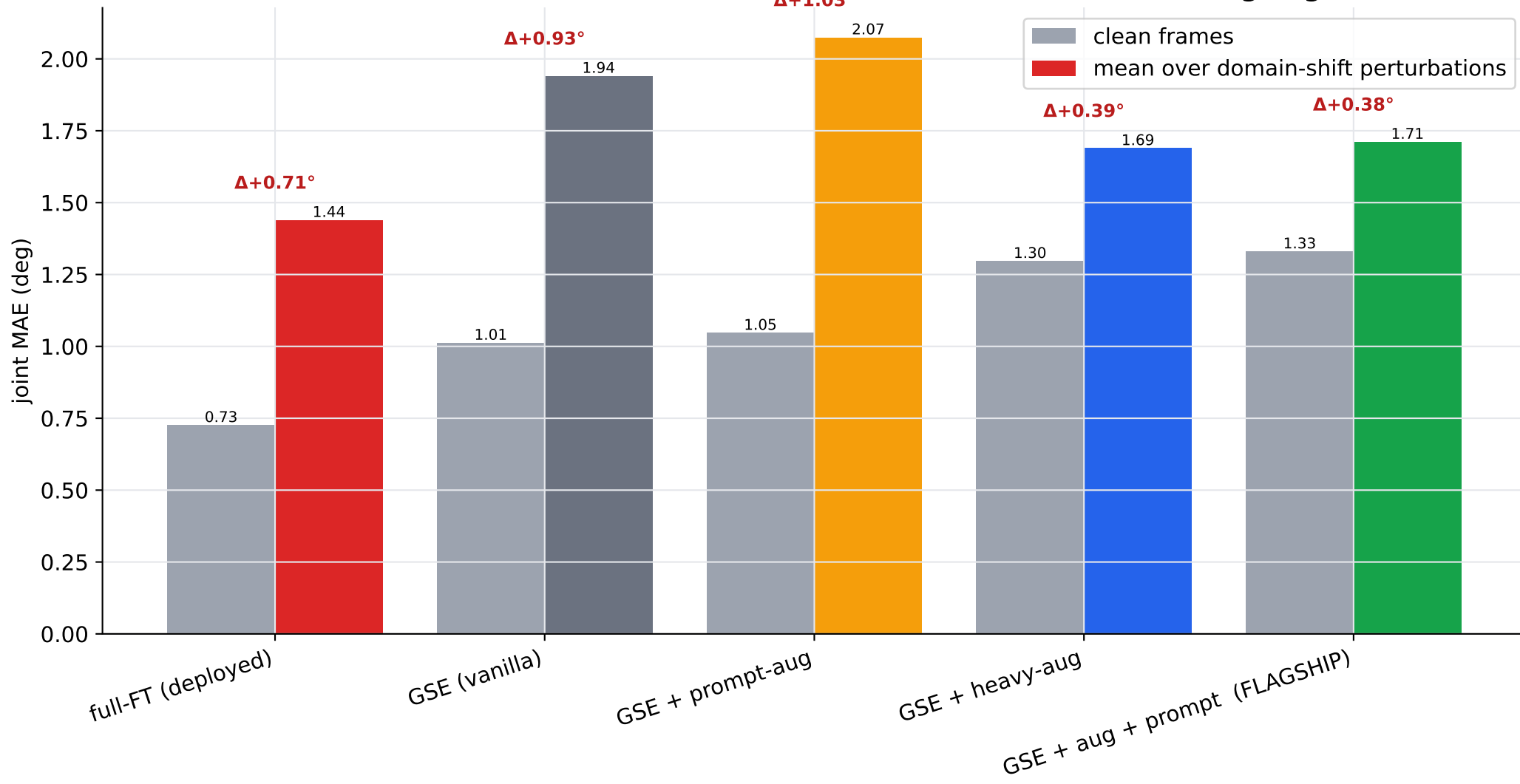


Robustness to domain shift — lower is better, esp. on the right-hand perturbations

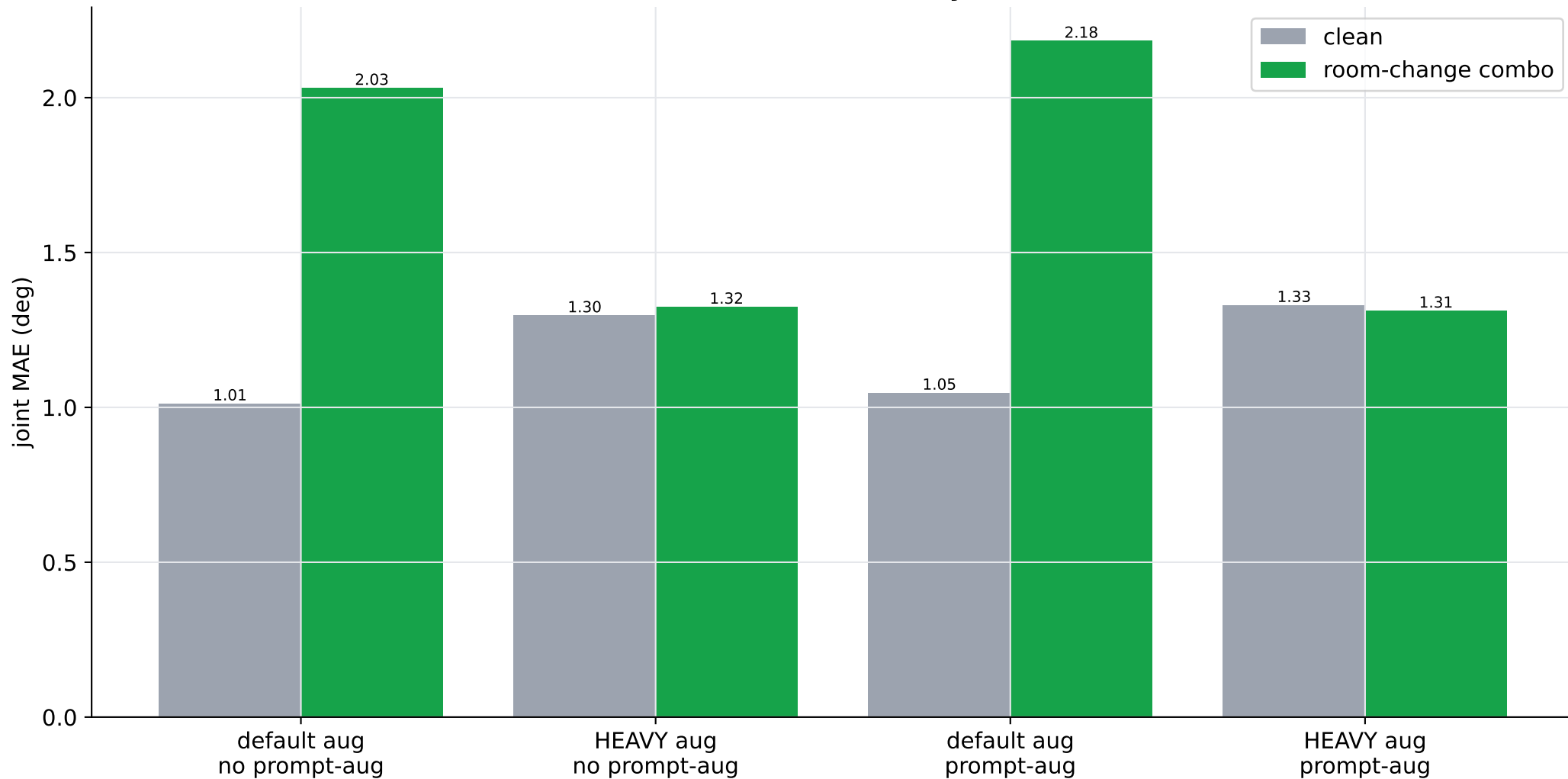


full-FT fits tighter on clean + mild shifts; the augmented GSE models degrade far less and win on the realistic 'room-change combo' (far right). Robustness is a FLAT profile, not just a low floor

Clean vs mean shifted error — the Δ increase IS the robustness measure (aug degrades least)



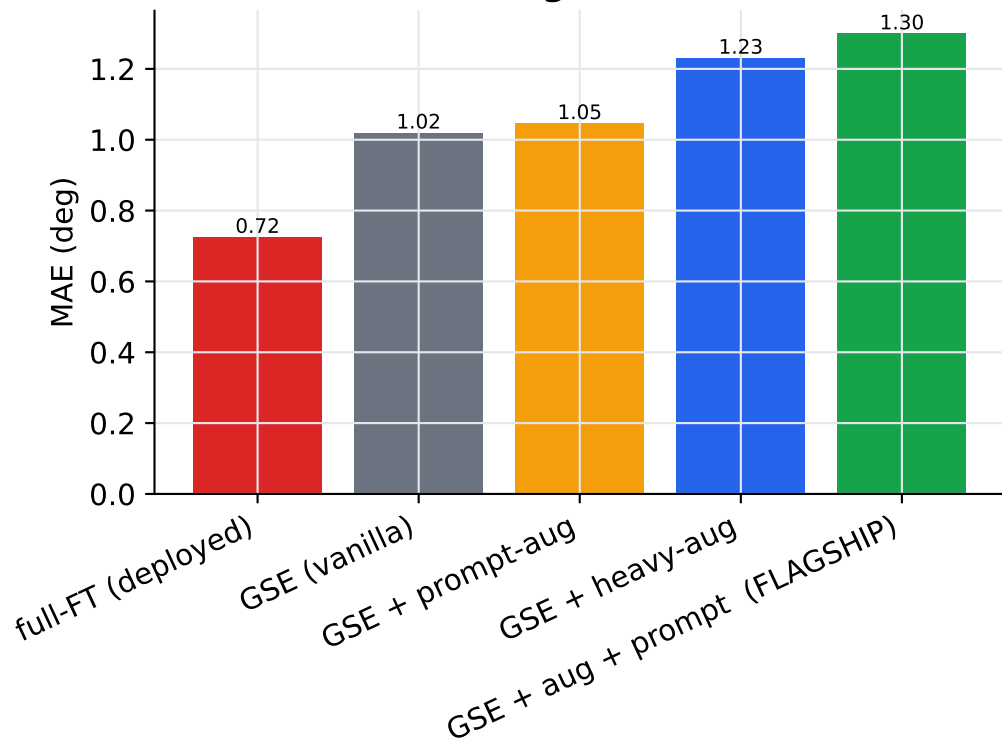
2×2 ablation (all GSE): which 'trick' buys domain robustness?



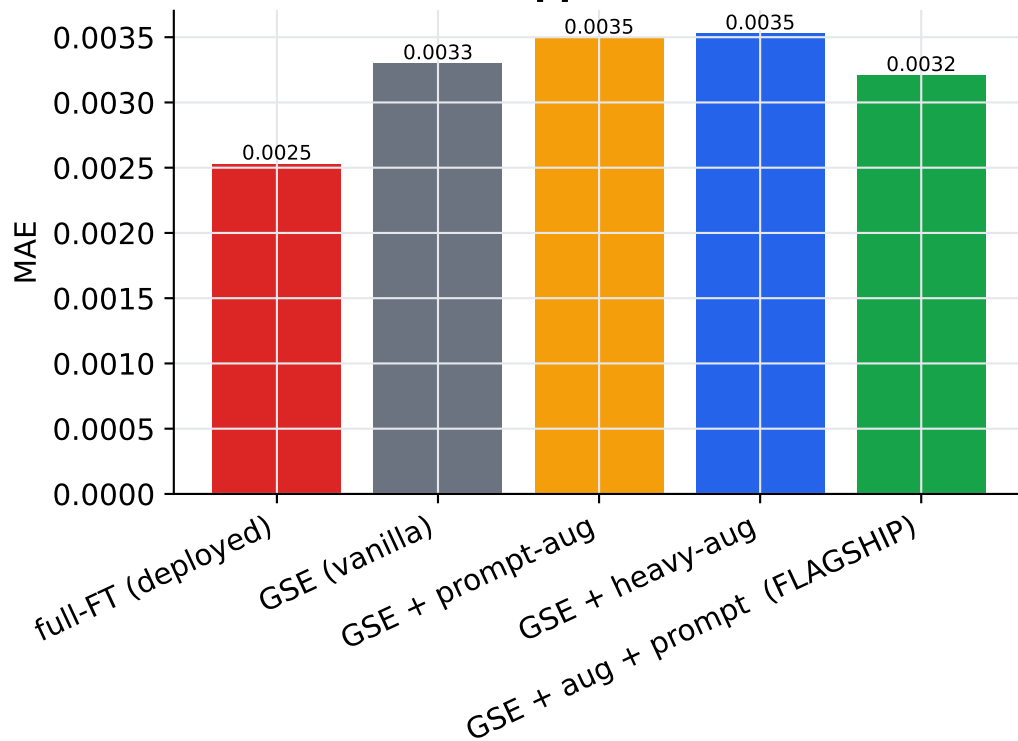
Compare the green (hard) bars across cells: heavy augmentation is the lever that shrinks the room-change error.

On the ORIGINAL frames everyone fits well — tight fit $\neq$ live robustness

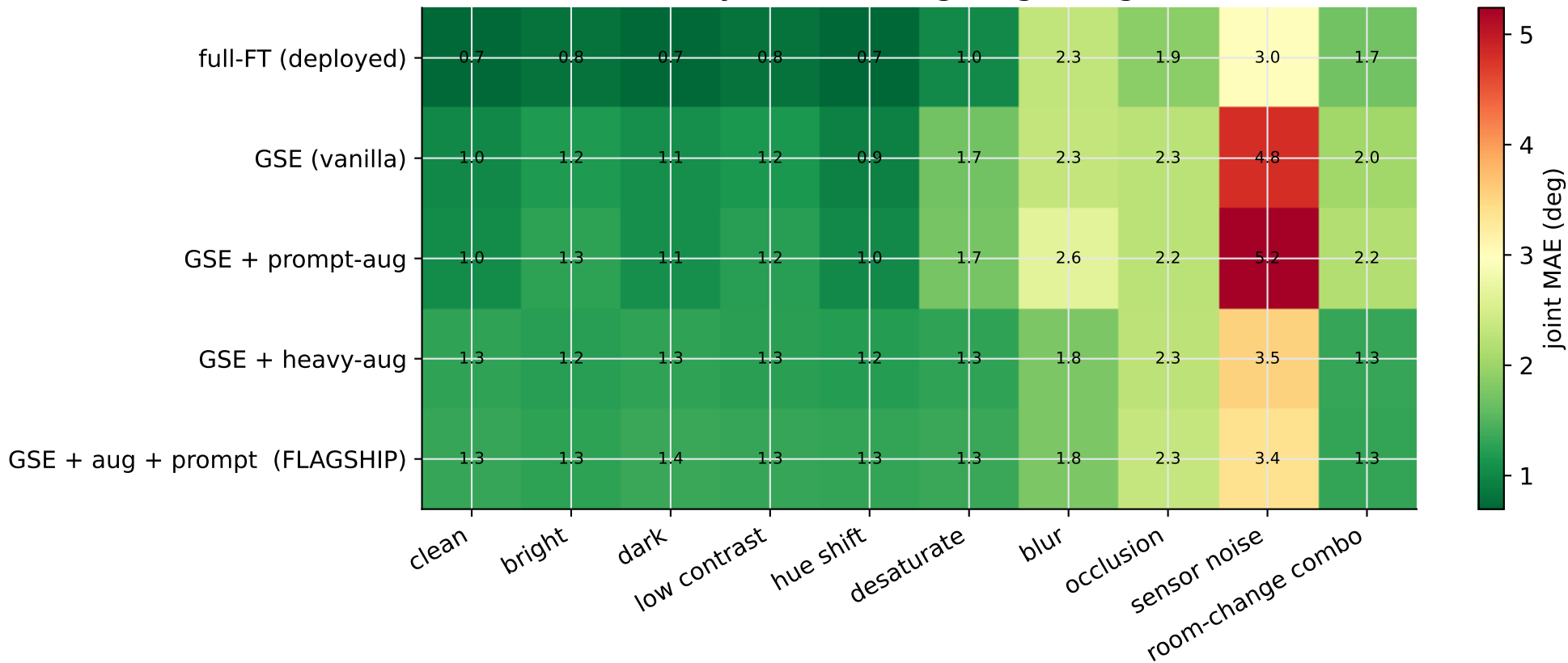
Next-step joint MAE (fit on original frames)



Gripper MAE

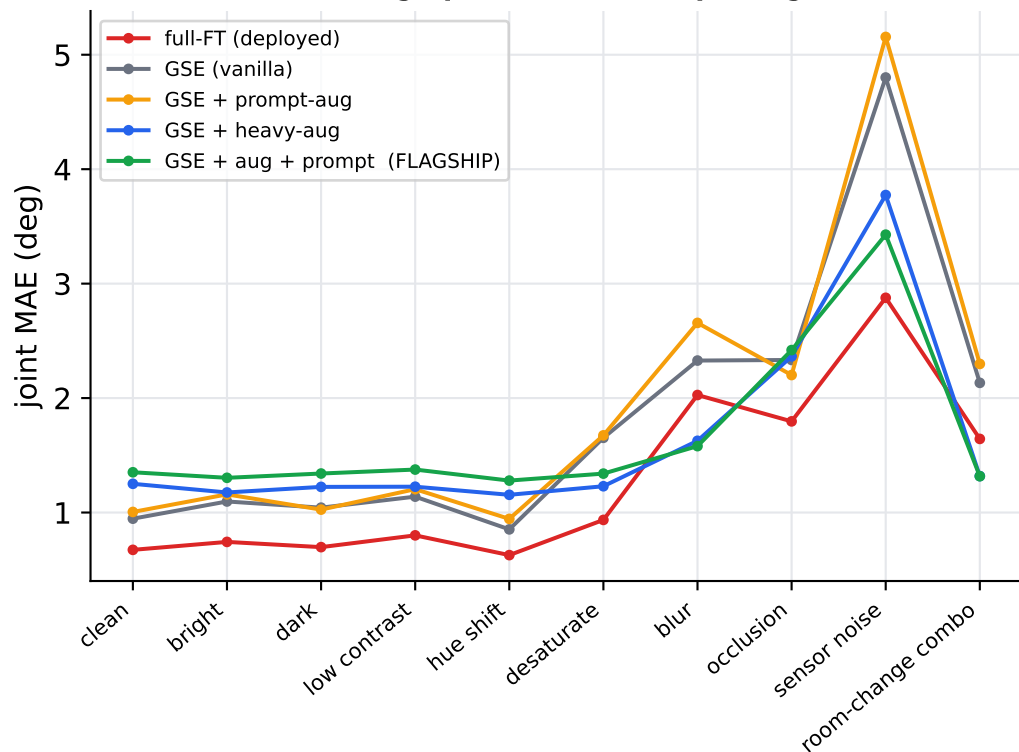


Per-condition joint MAE (deg) — green good, red bad

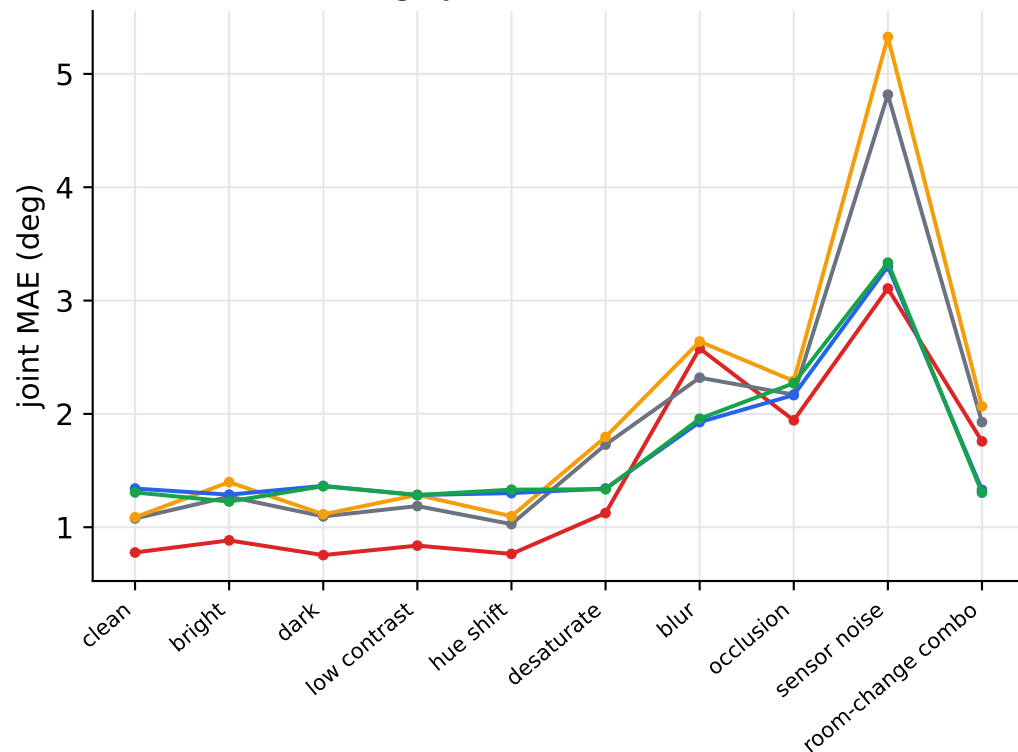


Per-task robustness across conditions

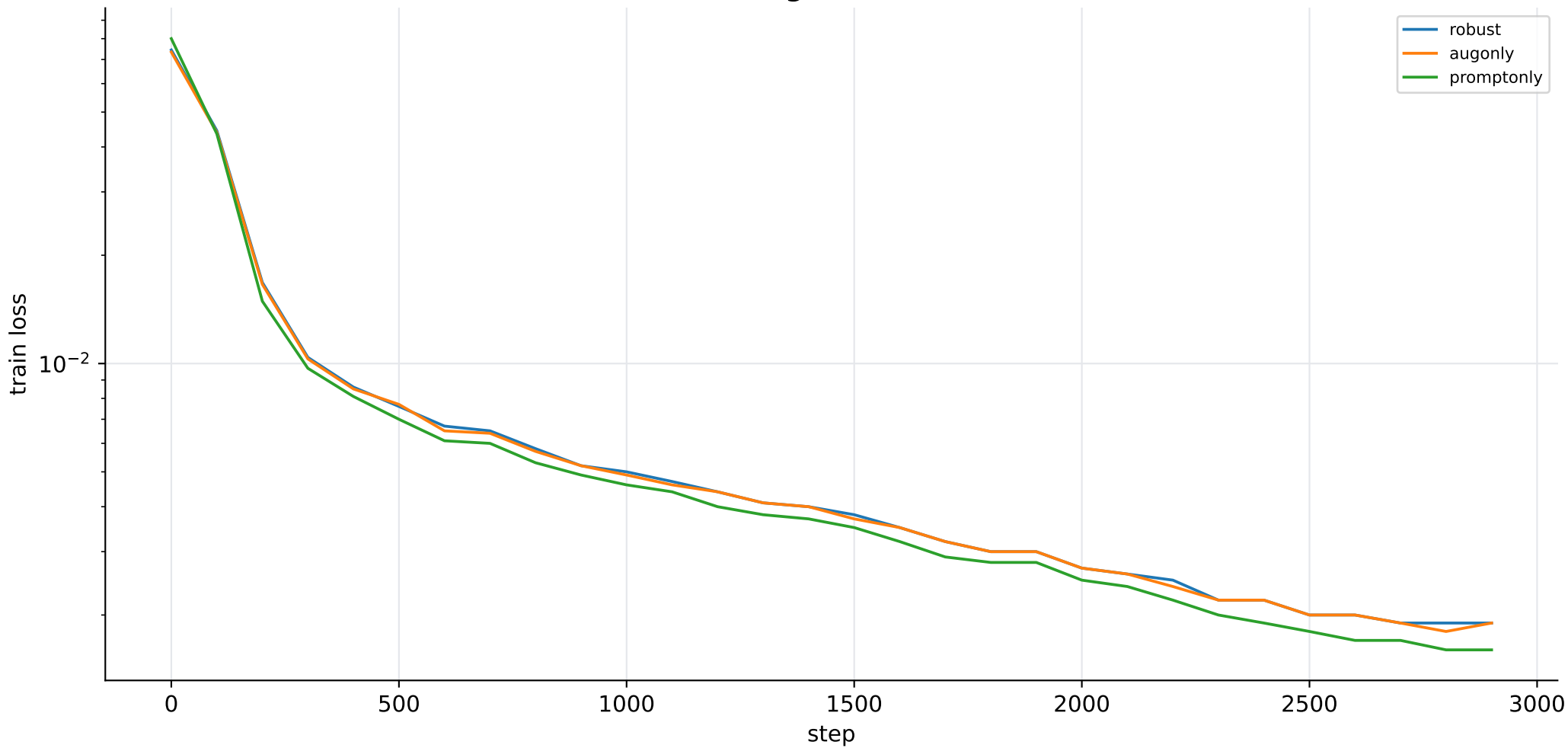
task: Picking up the bottle and placing it on



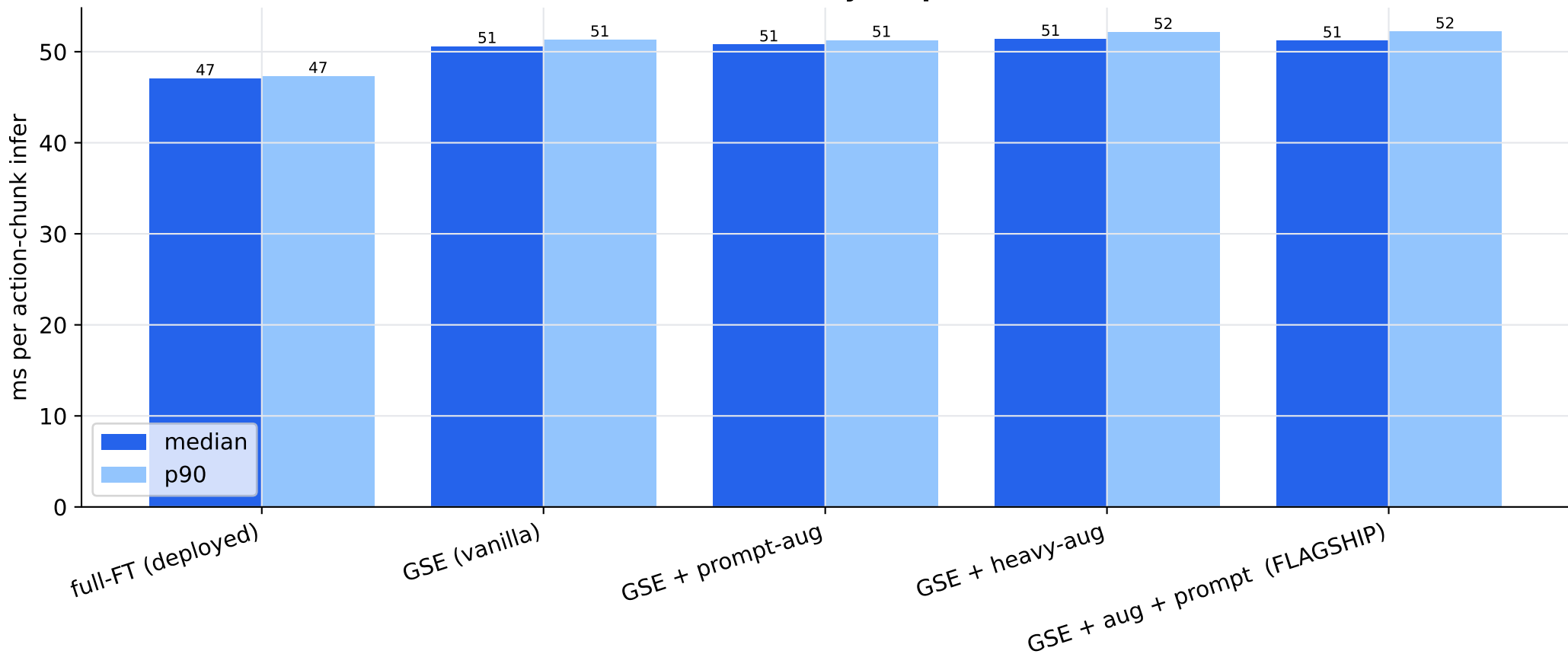
task: Picking up the bottle off of the box and



Training loss curves



Inference latency (in-process)



Conclusions & recommendation

WHAT WE DID Trained 3 GSE variants on 4xH100 (heavy domain-randomization and/or prompt paraphrasing) and compared them + the deployed full-FT + vanilla GSE, on clean frames and under synthetic domain-shift perturbations.

WHAT THE DATA SHOWS (honestly – it's not a clean sweep)

- Heavy visual augmentation is the lever for STABILITY. The 2x2 isolates it: adding it drops the room-change-combo error $\sim 2.0^\circ \rightarrow 1.3^\circ$ and nearly flattens degradation (clean $1.33^\circ \rightarrow$ combo 1.31°), vs the full FT's $0.73^\circ \rightarrow 1.70^\circ$.
- It is NOT a uniform win. The full FT's tight clean fit (0.73°) keeps it ahead on mild shifts and on perturbations the aug didn't cover (occlusion, heavy noise). Augmentation helps on what it trained on (colour / blur / combined room-change).
- Prompt paraphrasing slightly hurt the joint-fit metric and didn't improve visual robustness – its value (phrasing generalization, 2-task disambiguation) is real but NOT measured by this action-error eval.

RECOMMENDATION For the actual deploy problem (a different room $\approx$ the room-change combo), serve the flagship GSE+aug+prompt (GSE+aug is near-identical on the measured metric – either works):

```
SERVE_CONFIG=pi05_farm_uf850_gse SERVE_REPO=NoahWeiss/farm_uf850_pi05_gse_robust \
SERVE_STEP=2999 sbatch serve_pi05.sbatch
```

If you deploy in the ORIGINAL training room, the full FT is still tightest.

HONEST LIMITS / NEXT STEPS

- Synthetic perturbations are a PROXY; the real test is on-arm in the target room.
- Cheap wins: add occlusion + sensor-noise to the training aug (the conditions where aug didn't help), and collect a handful of in-room demos – in-domain data is the only thing that truly closes the gap.
- Inference is set up PI-style: RTC seam-smoothing cut chunk-to-chunk deviation 47-79% on the flagship. A/B it on the arm (RTC default vs --no-rtc).